

International Conference on Emerging Trends in Electrical and Mechanical Engineering

— Programme —

25 September 2026

Lisbon local time (UTC+1)

Morning programme

09:00–09:30	Opening session: Organising Committee of ETEME’26 Prof. Ana Veloso, Vice-Rector of the Polytechnic University of Coimbra (UPC), representing the Rector Prof. Nuno Cid Martins, President of the Coimbra Institute of Engineering (ISEC)		
09:30–10:00	Plenary session 1 Keynote Speaker: Rui Pestana (REN)		
10:00–10:15	Break and transition to parallel sessions		
10:15–12:00	Room 1 Track 1 – Energy, Power Systems and Electronics Chair: Paulo Moisés	Room 2 Track 2 – Mechanical Design, Materials, Manufacturing and Industrial Engineering Chair: Luís Serrano	Room 3 Track 3 – Mechatronics, Robotics and Automation Track 5 – Embedded Systems, Computational Models and Digital Twins Chair: Tomás Adão
	Low-Code Architecture for Behavioral Demand Response and Predictive Analytics (#05) ☐ R Artem Antonenko, Ievgen Zaitsev, Oleh Bondarenko	The Importance of Logistics, Supply Chain Management, and ABC Item Classification Optimization Through Cross-Departmental Collaboration (#08) ☐ I Edin Karisik	Experimental Investigation of TCP Rotation Influence on Robotic Sealant Deposition Along Curved Trajectories (#38) ☐ R ☐ S Fernando Benítez
	Analysis of the sustainability of the micro system approach method in the sizing of electrical conductors (#10) ☐ R Fátima Monteiro	Electro-Mechanical Experimental Assessment of a Variable Speed Centrifugal Pump: An Experimental Study for Industrial Systems (#12) ☐ I Raquel Faria	Development and Integration of an Embedded Electronics and Telemetry System for a Baja SAE Vehicle (#83) ☐ R ☐ S Henrique Soares Tartari
	Towards More Accurate Carbon Accounting in Power Systems: Incorporating Network Losses into Renewable Energy Assessments —Application to the Portuguese Electricity System (#15) ☐ R ☐ S António Duarte	Mechanical Characterisation of 3D-Printed Polylactic Acid Parts Reinforced with Different Loads of Cellulose Nanocrystals (#16) ☐ R ☐ S Maxim Rogozhkin	Adaptive Change Detection in Rotating Machinery Vibration Using Kernel CUSUM with a Stationarity-Gated Baseline (#33) ☐ R ☐ S Marco Aurélio Freitas Santos Júnior
	Smart Control of Hybrid Domestic Hot Water Production Based on Renewable Availability and Energy Prices (#17) ☐ R ☐ S Bruno Marques	Study on the Application of Denim Fibers in Thermoplastic Composites (#18) ☐ R ☐ S João Gomes	Unbalance Force Estimation in Rotors via Augmented Kalman Filter: A Trial-Weight-Free Approach (#41) ☐ R ☐ S Raimundo Neto
	Development of a Modular Photovoltaic Monitoring Platform for Laboratory-Based Energy Management Studies (#19) ☐ R Paulo Correia, Paulo Moisés Almeida da Costa	Sustainable manufacture of light alloy components based on the reuse of industrial waste (#21) ☐ R ☐ S Ricardo Costa	Rotating machine balancing via ANN: data Augmentation (#42) ☐ R ☐ S Raimundo Neto
	HVAC Modifications: Case Studies and Engineering Solutions (#27) ☐ I ☐ S Joana Veiga	Effect of path strategies on microstructure and mechanical properties in thin arc sections using wire arc additive manufacturing (#22) ☐ R Krishna Kishore Mugada	A Parametric Computational Framework for Aircraft Wetted Area Calculation and Zero-Lift Drag Estimation from 2D Layouts (#51) ☐ R ☐ S Felipe Dordan

*Each oral contribution consists of a 10-minute presentation followed by 5 minutes for questions. Contribution: ☐R Research; ☐I Industry; ☐S Student.

Afternoon programme

12:00–14:00	Lunch break		
14:00–14:15	Afternoon reopening		
14:15–14:45	Plenary session 2 Keynote Speaker: Marcelo Gaspar (University of Leiria and Oeste)		
14:45–15:00	Break and transition to parallel sessions		
15:00–16:15	Room 1 Track 1 – Energy, Power Systems and Electronics Chair: Dhaker Abbes	Room 2 Track 2 – Mechanical Design, Materials, Manufacturing and Industrial Engineering Chair: Pedro A. Marques	Room 3 Track 4 – Thermofluids, Heat Transfer and Thermal Systems Chair: João Malça
	Design of Electric Vehicle Charging Infrastructure for a Higher Education Institution: A Case Study (#49) ☐ R ☐ S Tatiana Miranda	Acoustic-Based Fault Identification in a Hair Dryer Using Machine Learning (#47) ☐ I ☐ S Francielle Souza, Diogo Lima, Stanley Washington Rezende, Bruno Barella	A web application for the preliminary selection of outdoor units in multi-split air conditioning systems (#03) ☐ I ☐ S Alexandre Furtado de Barros
	Human Centric Lighting in Classrooms: Visual Comfort, Energy Efficiency, and Sustainability (#50) ☐ R ☐ S André Rodrigues	ACM Framework™: An Integrated Methodology for Asset Criticality Assessment and Strategic Maintenance Decision-Making (#58) ☐ I Vânio Silva	A decision-support calculation tool based on a technical and economic analysis of the replacement of air-conditioning and DHW systems: a case study of the replacement of an LPG boiler (#04) ☐ I ☐ S Grigore Durbaca
	Economic Evaluation of a WtE Powerplant Integration into the District Heating System of Budapest (#59) ☐ R ☐ S Kende Kocsis	Design and Computational Structural Evaluation of a Baja SAE Tubular Spaceframe (#60) ☐ R ☐ S Gabriel Bianchini Matos	Energy Performance of a Variable Speed Centrifugal Fan for HVAC Applications (#13) ☐ R Anabela Carvalho
	Intelligent Fault Diagnosis in Boost Power Converters: A Systematic Literature Review (#72) ☐ R ☐ S Nuno Fidalgo	Sizing and Modeling of a 4x4 System Applied to a Baja SAE Vehicle (#61) ☐ I ☐ S Felipe Paiva	Energy Performance Assessment and Cost-Effective Retrofit Measures for a Hotel Building: A Portuguese Case Study (#14) ☐ R ☐ S Nuno Gomes
	Regulatory and Implementation Framework for Renewable Energy Communities in Portugal (#89) ☐ I ☐ S Tomás Adão		Influence of the operating point on the life cycle environmental impacts of a thermoelectric module for cooling applications (#20) ☐ R Mário Loureiro
16:15–16:30	Coffee break		
16:30–18:30	Room 1 Track 6 – Sustainability Assessment in Engineering Systems Chair: Adelino Pereira	Room 2 Track 2 – Mechanical Design, Materials, Manufacturing and Industrial Engineering Chair: Pedro Beirão	Room 3 Track 4 – Thermofluids, Heat Transfer and Thermal Systems Chair: Jesué Graciliano da Silva
	Engineering Framework for Improving Energy Efficiency in Existing Wastewater Treatment Plants (#11) ☐ R Stela Sefa	Characterization of the Ramp Milling Operation with Different Degrees of Inclination (#64) ☐ I Fernando Simões	Geographic Sensitivity of LFC-Based SHIP Systems Under Different Industrial Load Profiles (#25) ☐ I ☐ S Bernardo Farrero
	Comparative Life Cycle Environmental and Cost Assessment of Additively Manufactured and Conventionally Machined Mould Inserts for Injection Moulding (#43) ☐ I Pedro A. Marques	Optimizing Baja SAE Suspensions: A Kinematic and Structural Study (#70) ☐ R ☐ S Otávio Pinheiro	Physics-Informed Correction of RANS-Based CFD Fields for Backward-Facing Step Flow Using Hybrid Neural Networks (#31) ☐ R ☐ S Matheus Borges
	Comparative Life Cycle Costing and Environmental Assessment of Food-Contact LDPE Bags with Recycled Content (#44) ☐ I Pedro A. Marques	Development and Preliminary Validation of a MATLAB Application for Plantar Thermal Asymmetry Detection in Diabetic Foot Assessment (#71) ☐ R ☐ S Mónica Bastos	A conservative Euler–Lagrange conversion framework for primary atomisation: morphological classification and linear-stability fragmentation on adaptive parallel meshes (#37) ☐ R ☐ S Saulo Andrade Pinto
	Design of Electrical Infrastructure Solutions for a Data Center (#77) ☐ R ☐ S Luís Simões, Ricardo Fernandes	Development of a Low-Cost Sound Intensity Probe (#84) ☐ R ☐ S José Robson Oliveira Júnior	Computational Investigation of Integrated Battery Cooling Systems for Enhanced Thermal Management of EV Battery Packs (#67) ☐ R ☐ S Nancy Maurya
	Renewable energy-powered supply chain for enabling climate-resilient demand response for hard-to-abate industries (#68) ☐ R ☐ S KAVYA S	Environmental Anonymous Noise-Based In-Situ Sound Absorption Characterization Using Standard Laboratory Assets (#85) ☐ R ☐ S Daniel Lucas Coutinho da Silva	CFD Analysis of an Air Cooled 26650 Lithium-Ion Battery Pack with Flow Disturbing Cylindrical Fins (#69) ☐ I ☐ S Ashish Kumar Saini
	Technical and Economic Feasibility Analysis of Green Hydrogen Systems (#74) ☐ R ☐ S Miguel Machado, Nelson Luz	Toward a Bidirectional Vibration–Current Test Rig for Remaining Useful Life Prognostics of Rolling Bearings under Electrical Discharge Damage (#87) ☐ I ☐ S Gabriel Assunção Nicchetti, Diogo Lima, Francielle Souza, Eduardo Rabelo Domingues	Comparative Analysis of Pope and GRI-Mech Mechanisms for Methane Using the Cantera Library (#73) ☐ R ☐ S Maria Eduarda Nunes
	Less Energy, Greater Efficiency: The Future of Commercial Buildings (#75) ☐ R Paulo Vaz	Model-Assisted Design of an Alternative Autothermic Continuous Biochar Reactor for Agricultural Residue Valorization (#90) ☐ R ☐ S Frederico Kossack	CFD Modelling of Eucalyptus Woodchips Pyrolysis for Increased Renewable Hydrogen Production (#86) ☐ R ☐ S Mateus Dreveck Anziliero
	Integrated Life Cycle Costing and Environmental Assessment of Food-Contact Polypropylene Crates with Recycled Content (#45) ☐ I João Malça		
18:30–19:00	Awards and Distinctions		
19:00–19:15	Closing session		

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